Curriculum Vitae

Dr. Van Thuy Hoang

Personal Detail

Full name	Van Thuy Hoang (Mr.)
Date of Birth	20-01-1990
Nationality	Vietnamese
Current Address	Hanoi, Vietnam
Email	hoangvanthuy90@gmail.com
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Google scholar	https://scholar.google.com.vn/citations?user=E7Cc8rAAAAAJ

Research Interests

My research focuses on AI for Science (Bioinformatics) & Cybersecurity, developing Graph Foundation Models for scientific discovery and Cybersecurity. Key areas of interest include:

- **Foundation Models**: Pre-training & Adaptation strategies, Self-Supervised Learning.
- **Graph Learning**: Graph Structure Learning, Knowledge Graphs.
- **AI for Bioinformatics**: Molecular representation learning, molecular relational learning, molecular property prediction, and drug–drug interaction prediction.
- **Theoretical Machine Learning**: Information Theory, Causal Inference, Multi-view Learning, Explainable AI, Fairness Learning, Pattern Recognition.
- **AI for Cybersecurity**: Graph-based intrusion detection models.

Education

9/2022 –3/2026	Ph.D. in Artificial Intelligence, The Catholic University of Korea, South Korea.
2017 – 2018	M.S. in Information System, Le Quy Don Technical University, Vietnam.
2008 – 2013	B.S. in Information Technology, Le Quy Don Technical University, Vietnam.

Selected Publications *(Top-tier conferences/journals)*

- Van Thuy Hoang, O-Joun Lee. *Pre-training Graph Neural Networks on Molecules by using Subgraph-conditioned Graph Information Bottleneck.* In Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI 2025) **(Ranking A*)**
- Van Thuy Hoang, Hyeon-Ju Jeon, O-Joun Lee. *Mitigating Degree Bias in Graph Representation Learning with Learnable Structural Augmentation and Structural Self-attention.* IEEE Transactions on Network Science and Engineering (IEEE TNSE 2025) (**Top 1.8% JCR**)
- Van Thuy Hoang, O-Joun Lee. *Transitivity-Preserving Graph Representation Learning for Bridging Local Connectivity and Role-based Similarity.* In Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI 2024) (**Ranking A***)
- Nam-Gyu Jung, Van Thuy Hoang, O-Joun Lee, Chang Choi. Kiosk Recommend System based on Self-Supervised Representation Learning of User Behaviors in Offline Retail. IEEE Internet of Things Journal, February 2024. **(Top 3.4% JCR)**

Full Publications

2026:

1. Van Thuy Hoang, O-Joun Lee: Context-aware Graph Causality Inference for Few-Shot Molecular Property Prediction. arXiv preprint arXiv:2601.11135. *Under review.*
2. Van Thuy Hoang, O-Joun Lee: Pre-training Graph Neural Networks on 2D and 3D Molecular Structures by using Multi-View Conditional Information Bottleneck. arXiv preprint arXiv:2511.18404. *Under review.*

2025:

3. Van Thuy Hoang, O-Joun Lee. Pre-training Graph Neural Networks on Molecules by using Subgraph-conditioned Graph Information Bottleneck. In Proceedings of the 39th AAAI Conference on Artificial Intelligence 2025. AAAI/ISSN: 21595399, DOI: 10.1609/aaai.v39i16.33891 , CORE A*, Scopus
4. Van Thuy Hoang, Hyeon-Ju Jeon, O-Joun Lee. Mitigating Degree Bias in Graph Representation Learning with Learnable Structural Augmentation and Structural Self-attention. IEEE Transactions on Network Science & Engineering 2025. IEEE/ISSN: 2327-4697, DOI: 10.1109/TNSE.2025.3563697, SCIE Q1, IF = 7.3. (JCR 2025, Top 2.6%).
5. Van Thuy Hoang, Tien-Bach-Thanh Do, Jinho Seo, Seung Charlie Kim, Luong Vuong Nguyen, Duong Nguyen Minh Huy, Hyeon-Ju Jeon, O-Joun Lee. Halal or Not: Knowledge Graph Completion for Predicting Cultural Appropriateness of Daily Products. IEEE Access, 2025. IEEE/ ISSN: 2169-3536, DOI: 10.1109/ACCESS.2025.3528488, SCIE Q2 (JCR 2025). IF=4.2
6. Thien Thi Thu Nguyen, Eun-Soon You, Hyun Woo Kim, Van Thuy Hoang, Luong Vuong Nguyen, O-Joun Lee. Constructing a Companion Animal Disease Knowledge Graph by Utilizing LLMs in Data Preprocessing and Pseudo Annotation. Proceedings of the International Conference on Research in Adaptive and Convergent Systems (RACS 2025), ACM/ISBN: 979-840072231-8, DOI: 10.1145/3769002.3769990 , Scopus

2024:

7. Van Thuy Hoang, O-Joun Lee. Transitivity-Preserving Graph Representation Learning for Bridging Local Connectivity and Role-based Similarity. In Proceedings of the 38th AAAI Conference on Artificial Intelligence 2024, AAAI/ISSN: 21595399, DOI: 10.1609/aaai.v38i11.29138 , CORE A*, Scopus
8. Nam-Gyu Jung, Van Thuy Hoang, O-Joun Lee, Chang Choi. Kiosk Recommend System based on Self-Supervised Representation Learning of User Behaviors in Offline Retail. IEEE Internet of Things Journal, February 2024. SCIE Q1, IF = 8.9. DOI: 10.1109/JIOT.2024.3365144 , IEEE/ISSN: 2327-4662.
9. Van Thuy Hoang, O-Joun Lee. An Overview of Pre-Trained Graph Models for Molecular Structure Analysis. In IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia 2024). IEEE/ISBN: 979-833153083-9, DOI: 10.1109/ICCE-Asia63397.2024.10773928 , Scopus

2023:

10. Van Thuy Hoang, T. S. Nguyen, S. Lee, J. Lee, L. V. Nguyen, O.-J. Lee. Companion Animal Disease Diagnostics Based on Literal-Aware Medical Knowledge Graph Representation Learning. IEEE Access, 2023. IEEE/ ISSN: 2169-3536. DOI: 10.1109/ACCESS.2023.3324046 SCIE, Q2, IF = 3.9.
11. Van Thuy Hoang, Hyeon-Ju Jeon, Eun-Soon You, Y.W. Yoon, Sungyeop Jung, O-Joun Lee. Graph Representation Learning and Its Applications: A Survey. Sensors, 2023. SCIE Q2, IF = 3.4. DOI: 10.3390/s23084168 . ISSN: 1424-8220

2018:

12. Van Thuy Hoang, Phan Viet Anh, and Nguyen Xuan Hoai. "Automated large program repair based on big code" In Proceedings of the 9th International Symposium on Information and Communication Technology 2018 (ISICT 2018). ACM/ISBN: 978-145036539-0, DOI: 10.1145/3287921.3287958 , Scopus

13. Ly Vu, Van Thuy Hoang, Uy Nguyen, Tran Ngoc, Diep Nguyen, Hoang Dinh, Eryk Dutkiewicz. Time Series Analysis for Encrypted Traffic Classification: A Deep Learning Approach. In Proceedings of the 18th International Symposium on Communications and Information Technologies 2018 (ISCIT 2018). IEEE/ISBN: 978-153868458-0, DOI: 10.1109/ISCIT.2018.8587975 , Scopus

2013:

14. Loi Cao Van, Van Thuy Hoang, Quang Uy Nguyen. A scheme for building a dataset for intrusion detection systems. In Proceeding of the Third World Congress on Information and Communication Technologies (WICT), IEEE/ISBN:978-1-4799-3230-6 DOI: 10.1109/WICT.2013.7113149 , Scopus

Patents

1. O-Joun Lee, Van Thuy Hoang: System for predicting molecular property and method thereof (분자 특성 예측 시스템 및 그 방법). Ref. No: KR 1020260008863, 01/2026
2. O-Joun Lee, Van Thuy Hoang: Method for pre-training graph neural network for 2D and 3D molecular structures and system thereof (분자의 2차원 및 3차원 구조에 대한 그래프 신경망 사전학습 방법 및 그 시스템). Ref. No: KR 1020250169745, 11/2025
3. O-Joun Lee, Van Thuy Hoang: Method for pre-training using graph information bottleneck based on subgraph and system thereof (서브그래프 기반 그래프 정보 병목 기법을 이용한 사전학습 방법 및 그 시스템). Ref. No: KR 1020250030478, 03/2025
4. O-Joun Lee, Van Thuy Hoang, Hochan Yang, Ju Hee Shim, Seung Kim, SungGyu Kim: System of Recommending Halal Cosmetics based on Knowledge Graph Representation Learning (지식 그래프 표현 학습에 기반한 할랄 화장품 추천 시스템). Ref. No: KR 1020240159542, 11/2024
5. O-Joun Lee, Van Thuy Hoang: Graph Represntation Learning Method and System Thereof (그래프 표현 학습 방법 및 그 시스템). Ref. No: KR 1020240104783, 08/2024
6. O-Joun Lee, Van Thuy Hoang, Sang Thanh Nguyen, Dong Kook Park, Doyul Lee: Apparatus and Method for Representation Learning of Knowledge Graph using Heterogeneous Data Comprehensively (이종 데이터를 통합적으로 이용하는 지식 그래프 표현 학습 방법 및 장치). Ref. No: KR 1020240090256, 07/2024
7. O-Joun Lee, Van Thuy Hoang: Apparatus and Method for Training Graph Neural Network based on Graph Augmentation (그래프 증강 기반 그래프 뉴럴 네트워크 학습 방법 및 장치). Ref. No: KR 1020240086072, 07/2024
8. O-Joun Lee, Van Thuy Hoang, Ho Beom Kim, Jinho Seo, Seung Charlie Kim, Nox Kim: System and Method for Predicting Availability based on Knowledge Graph (지식 그래프 기반의 가용성 예측 시스템 및 방법). Ref. No: KR 1020230171167, 11/2023

Academic Services

- Conference Reviewer: AAAI-26, ICLR-26, NeurIPS-26, AAAI-27, ICLR-27
- Journal Reviewer: Artificial Intelligence Review, Pattern Recognition, Journal of Big Data, Neurocomputing, Knowledge-Based Systems, Expert Systems with Applications, npj Computational Materials, Information Sciences.

Preprint

- Van Thuy Hoang, O-Joun Lee: A Survey on Structure-Preserving Graph Transformers. 01/2024. arXiv preprint. arXiv:2401.16176.
- Thanh Sang Nguyen, Jooho Lee, Van Thuy Hoang, O-Joun Lee: Connector 0.5: A unified framework for graph representation learning. 04/2023. arXiv preprint. arXiv:2304.13195

References

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